

Thermal Management & Fluid Dynamics Validation Report

Empirical Chiller Testing, PTC Response, and Theoretical Simulations

1. Battery Pack Coolant Chiller Closed-Loop Regulation

- Test Case TC-THM-041: Thermal bench test: inlet coolant temperature stabilized at 25.4°C under 5.2 kW steady thermal dissipation. V
- Test Case TC-THM-042: Compressor speed sweep from 1000 rpm to 8500 rpm verified; maximum observed speed error was 28 rpm.

2. Cabin Heat Pump Thermal Capacity & EXV Modulation

- Test Case TC-THM-043: Heat pump tested at 0°C and -5°C ambient delivering 5.1 kW and 4.8 kW respectively. Extreme cold chamber
- Test Case TC-THM-044: EXV positioning verified at +25°C ambient across full 500 steps. Cold step validation at -30°C scheduled for n

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3. PTC Cabin Heater Transient Response

- Test Case TC-THM-046: PTC heater step response test: measured 32.4 seconds to reach 90% power (5.4 kW) due to soft-start inrush

4. Analytical Calculations & Simulation Records

- Venting Acoustic Modeling: Acoustic sensor baseline signal characterized in bench chamber. Sensor response demonstrated under ge
- Inverter Current THD Model: MATLAB/Simulink power stage simulation predicts current THD of 2.4% under ideal switching parameters
- DC-DC Surge Clamp Model: SPICE circuit simulation shows clamping TVS diode limits pulse 2a peak to 16.2 V.
- Grid Surge Varistor Model: Transient surge suppression modeled in LTspice with MOV + gas tube surge arrestor.
- Comfort PMV Calculation: CFD thermal cabin comfort modeling demonstrates PMV of +0.2 under simulated 22°C ambient.
- NVRAM Wear Calculation: NVRAM endurance calculation model estimates 125,000 cycles based on Flash memory manufacturer cell
- Moisture Permeation Model: Conformal coating moisture permeation model calculates insulation resistance > 100 M· under 95% RH.